

**Mass dimension: 4**

**nD: 0, H: 2, H+: 2**

**SU2\_W-Basis Tensors**

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (1)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (2)$$

**H0H0H+0H+0**

$$-\left(\frac{1}{2}\hat{\mu}_h^2\right)(T_{SU2})_{ijkl}^0 H_i H_j (H^\dagger)_k (H^\dagger)_l \quad (3)$$

$$+\left(\frac{1}{4}\hat{\mu}_h^2\right)(T_{SU2})_{ijkl}^1 H_i H_j (H^\dagger)_k (H^\dagger)_l \quad (4)$$

**Mass dimension: 6**

**nD: 2, H: 2, H+: 2**

**SU2\_W-Basis Tensors**

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (5)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (6)$$

**H0H0H+1H+1**

$$-\left(\frac{3}{4}\right)M_s^{-2}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(T_{SU2})_{ijkl}^0 H_i H_j D_{\alpha_1}^{\dot{\alpha}_1}(H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_l \quad (7)$$

$$+\left(\frac{5}{8}\right)M_s^{-2}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(T_{SU2})_{ijkl}^1 H_i H_j D_{\alpha_1}^{\dot{\alpha}_1}(H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_l \quad (8)$$

**H0H1H+0H+1**

$$+\left(\frac{1}{2}\right)M_s^{-2}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(T_{SU2})_{ijkl}^1 H_i D_{\alpha_1}^{\dot{\alpha}_1} H_j (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_l \quad (9)$$

**nD: 0, H: 3, H+: 3**

**SU2\_W-Basis Tensors**

$$(T_{SU2})_{ijklmn}^0 = \varepsilon_{il}\varepsilon_{jm}\varepsilon_{kn} \quad (10)$$

$$(T_{SU2})_{ijklmn}^1 = \varepsilon_{ik}\varepsilon_{jm}\varepsilon_{ln} \quad (11)$$

$$(T_{SU2})_{ijklmn}^2 = \varepsilon_{ik}\varepsilon_{jl}\varepsilon_{mn} \quad (12)$$

$$(T_{SU2})_{ijklmn}^3 = \varepsilon_{ij}\varepsilon_{km}\varepsilon_{ln} \quad (13)$$

$$(T_{SU2})_{ijklmn}^4 = \varepsilon_{ij}\varepsilon_{kl}\varepsilon_{mn} \quad (14)$$

**H0H0H0H+0H+0H+0**

$$-\left(\frac{1}{2}\lambda_h\right)M_s^{-2}(T_{SU2})_{ijklmn}^0H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (15)$$

$$+\left(\frac{1}{6}\lambda_h\right)M_s^{-2}(T_{SU2})_{ijklmn}^1H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (16)$$

$$+\left(\frac{1}{6}\lambda_h\right)M_s^{-2}(T_{SU2})_{ijklmn}^2H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (17)$$

$$+\left(\frac{1}{6}\lambda_h\right)M_s^{-2}(T_{SU2})_{ijklmn}^3H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (18)$$

$$-\left(\frac{1}{3}\lambda_h\right)M_s^{-2}(T_{SU2})_{ijklmn}^4H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (19)$$

**nD: 0, H: 1, H+: 2, Q+: 1, uC+: 1**

**SU3\_C-Basis Tensors**

$$(T_{SU3})_{abc}^0 = +\left(\frac{1}{2}\right)\varepsilon_{abc} - \left(\frac{1}{2}\right)\varepsilon_{bac} \quad (20)$$

**SU2\_W-Basis Tensors**

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (21)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (22)$$

**H0H+0H+0Q+0uC+0**

$$+\left(\frac{1}{8}\right)M_s^{-2}(y_u)^{uv}\varepsilon_{\dot{\alpha}2\dot{\alpha}1}(T_{SU2})_{ijkl}^0(T_{SU3})_{abc}^0H_i(H^\dagger)_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}1,u}(u_\mathbb{C}^\dagger)_c^{\dot{\alpha}2,v} \quad (23)$$

$$+\left(\frac{1}{8}\right)M_s^{-2}(y_u)^{uv}\varepsilon_{\dot{\alpha}2\dot{\alpha}1}(T_{SU2})_{ijkl}^1(T_{SU3})_{abc}^0H_i(H^\dagger)_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}1,u}(u_\mathbb{C}^\dagger)_c^{\dot{\alpha}2,v} \quad (24)$$

**nD: 0, L: 1, eC: 1, H: 1, H+: 2**

**SU2\_W-Basis Tensors**

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (25)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (26)$$

**L0eC0H0H+0H+0**

$$-\left(\frac{1}{2}\right)M_s^{-2}(y_e^\dagger)^{vu}\varepsilon^{\alpha_1\alpha_2}(T_{SU2})_{ijkl}^0L_{\alpha_1,i}^u(e_\mathbb{C})_{\alpha_2}^vH_j(H^\dagger)_k(H^\dagger)_l \quad (27)$$

$$+\left(\frac{1}{4}\right)M_s^{-2}(y_e^\dagger)^{vu}\varepsilon^{\alpha_1\alpha_2}(T_{SU2})_{ijkl}^1L_{\alpha_1,i}^u(e_\mathbb{C})_{\alpha_2}^vH_j(H^\dagger)_k(H^\dagger)_l \quad (28)$$

**nD: 0, Q: 1, dC: 1, H: 1, H+: 2**

**SU3\_ C-Basis Tensors**

$$(T_{SU3})_{abc}^0 = +\left(\frac{1}{2}\right)\varepsilon_{abc} - \left(\frac{1}{2}\right)\varepsilon_{acb} \quad (29)$$

**SU2\_ W-Basis Tensors**

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (30)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (31)$$

**Q0dC0H0H+0H+0**

$$-\left(\frac{1}{4}\right)M_s^{-2}(y_d^\dagger)^{vu}\varepsilon^{\alpha_1\alpha_2}(T_{SU2})_{ijkl}^0(T_{SU3})_{abc}^0Q_{\alpha_1,i,a}^u(d_{\mathbb{C}})_{\alpha_2,bc}^vH_j(H^\dagger)_k(H^\dagger)_l \quad (32)$$

$$+\left(\frac{1}{8}\right)M_s^{-2}(y_d^\dagger)^{vu}\varepsilon^{\alpha_1\alpha_2}(T_{SU2})_{ijkl}^1(T_{SU3})_{abc}^0Q_{\alpha_1,i,a}^u(d_{\mathbb{C}})_{\alpha_2,bc}^vH_j(H^\dagger)_k(H^\dagger)_l \quad (33)$$